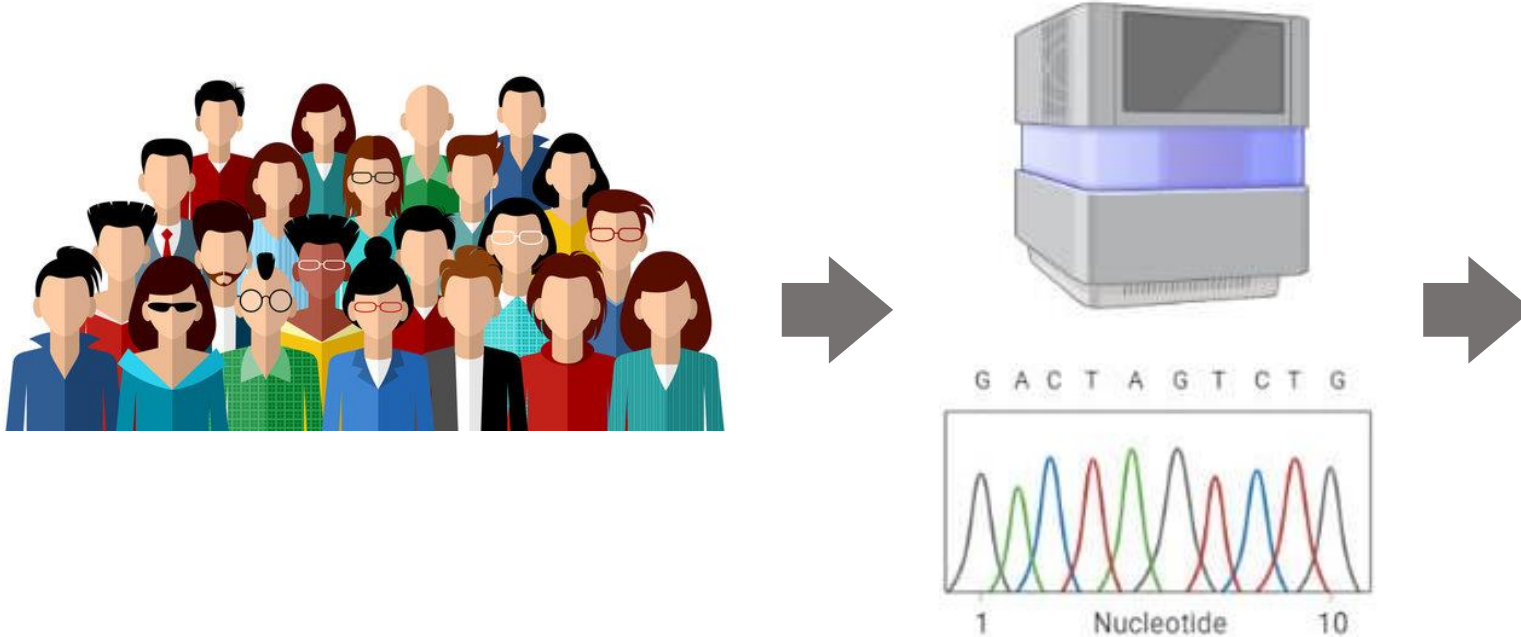


Big Data Analysis

Ahmed Mahfouz

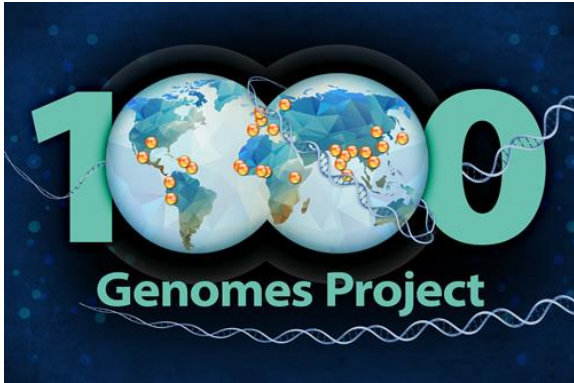
Dept. of Human Genetics, Leiden University Medical Center
Pattern Recognition and Bioinformatics, TU Delft

The promise of genomic medicine

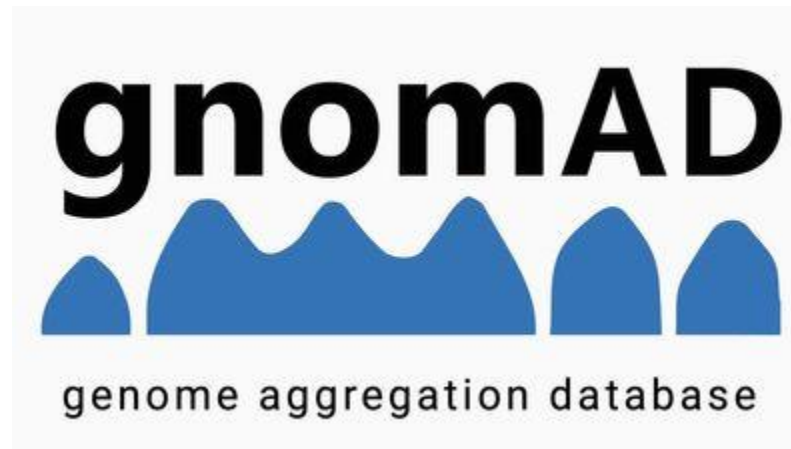
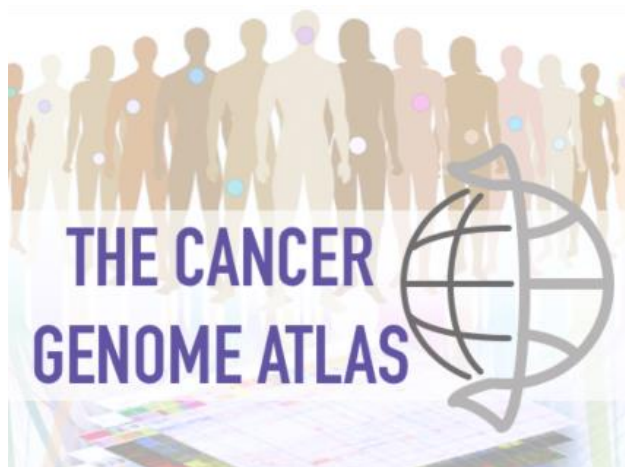


- Disease mechanisms
- New target genes
- New therapeutics
- Personalized medicine

Creation of Variant Catalogs

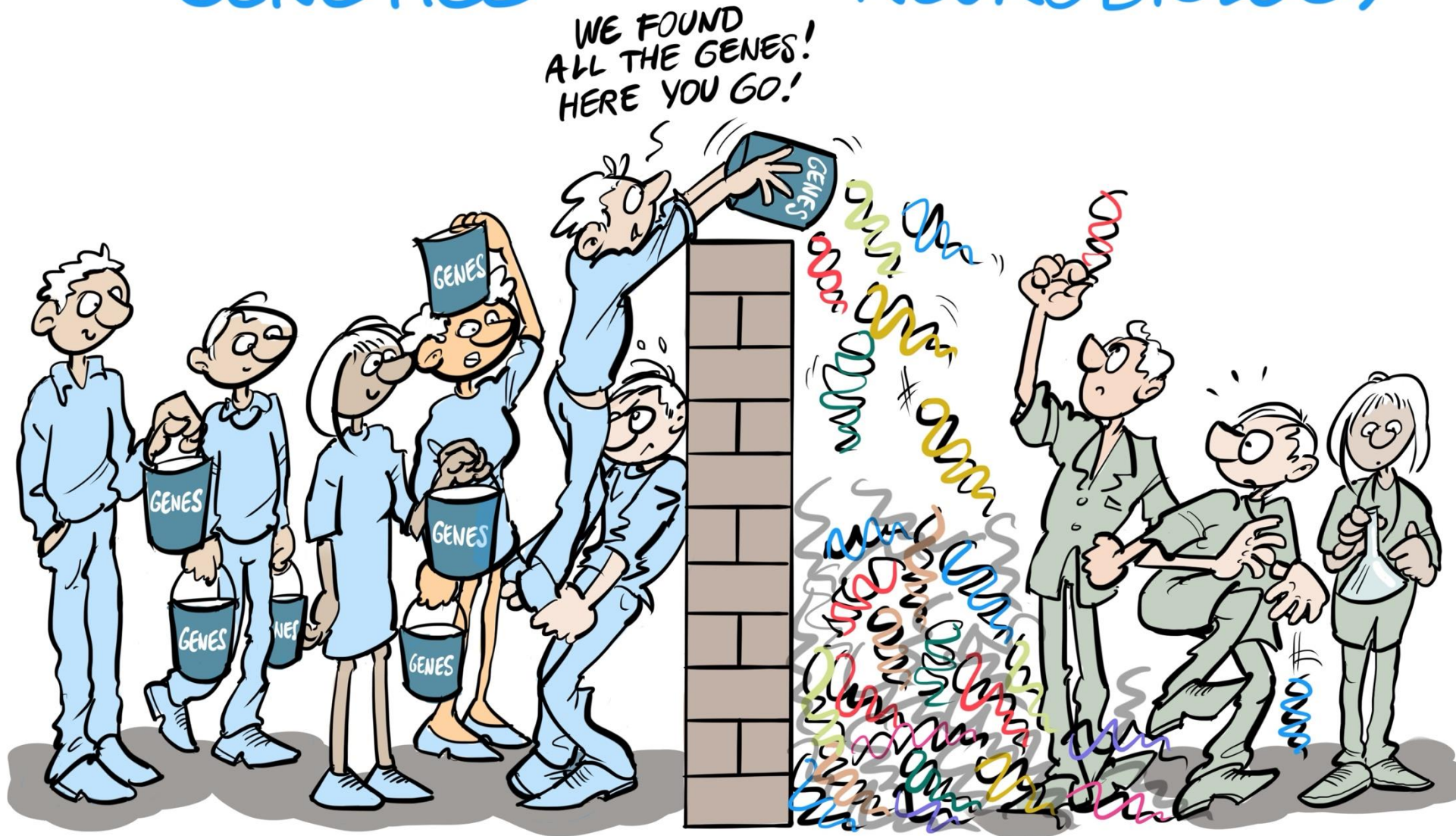


THE PRECISION MEDICINE INITIATIVE

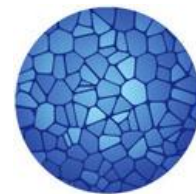
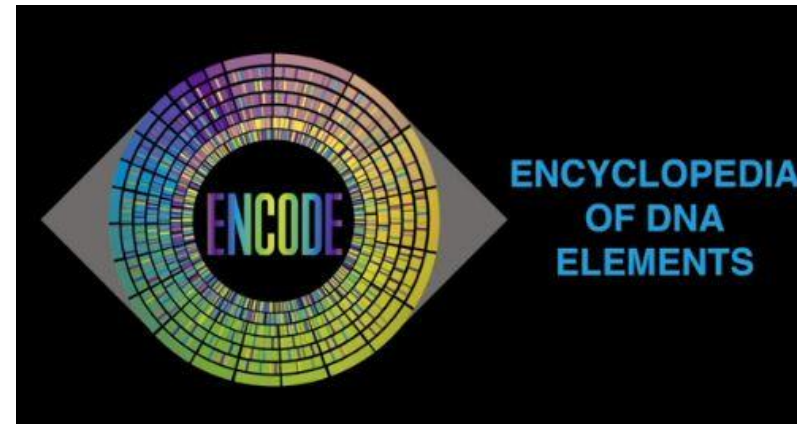
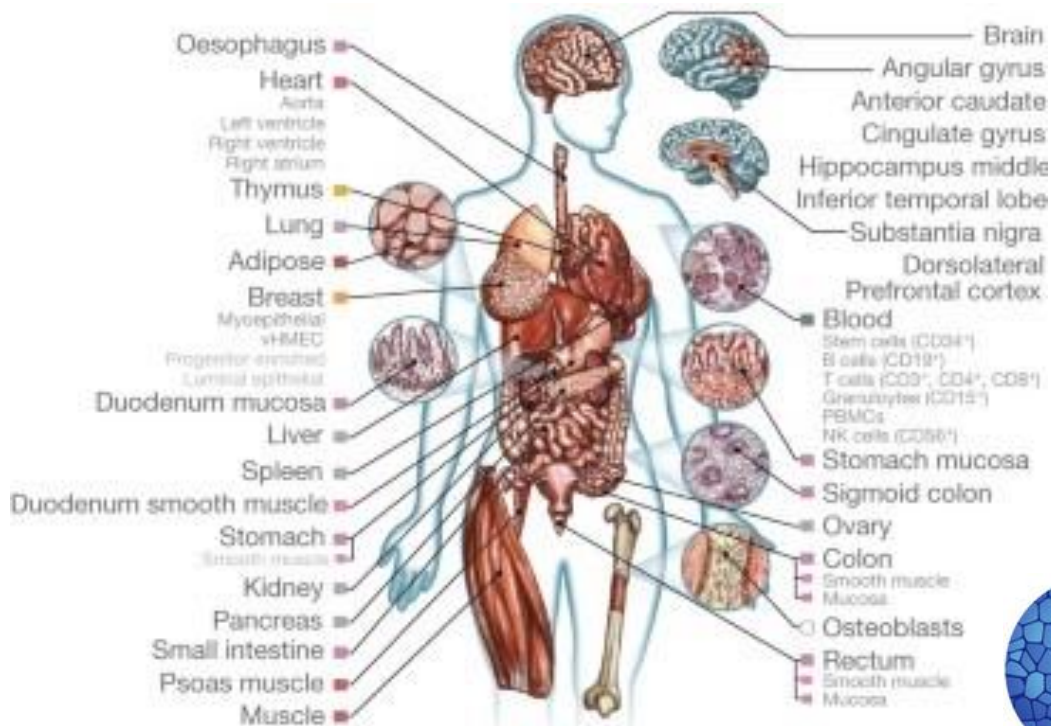


GENETICS

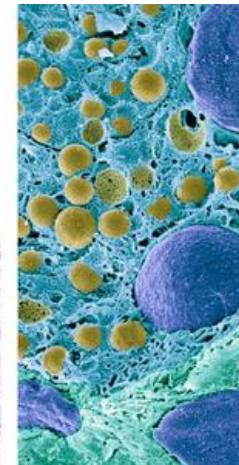
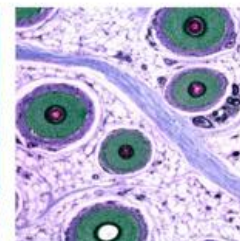
NEUROBIOLOGY



Resources for Variant Interpretation



HUMAN
CELL
ATLAS



Big Data Analysis

- Lecture 1
 - Linking variants to genes
 - Linking genes to functions
- Lecture 2
 - Using AI to model genomic sequences
 - Using AI models to predict variant effects?
- Lecture 3
 - Single cell genomics
 - Single cell data analysis
- Lecture 4
 - Functional genomics using single-cell sequencing

Do it yourself: mini-project

Schedule

Monday 12 January

9:15 – 9:30	Introduction
9:30 – 10:15	Variant to gene to function
10:15 – 10:30	Break
10:30 – 11:15	AI for genomics
11:15 – 12:30	Work on mini-project
12:30 – 13:30	Lunch
13:30 – 17:30	Work on mini-project

Wednesday 14 January

9:15 –10:15	Single-cell genomics
10:15 – 10:30	Break
10:30 –11:30	Functional genomics using single-cell RNA-seq
11:30 – 12:30	Work on mini-project
12:30 – 13:30	lunch
13:30 – 15:30	Work on mini-project (hand-in before 15:30)